

Radiative cooling of a helium-dominated wind

Abstract

A helium-dominated escaping atmosphere has almost no line cooling. Every line channel available to helium has a high excitation threshold (19.8 eV for the lowest, against 10.2 eV for Lyman- α), so the Boltzmann factor suppresses helium line cooling by five to six orders of magnitude relative to Lyman- α at 10^4 K. The energy budget of such a wind therefore closes on continuum channels and adiabatic expansion rather than on lines, and the gas has no effective thermostat. This memo records the channels implemented in `Cool_coeff.f90`, their measured relative sizes, why the He I 584 Å resonance line is already inside the existing coefficient even though no term carries its threshold, and what Cherubim et al. (2026) do and do not say about any of this.

1. Why the question arises

The metastable helium detection at LHS 1140b [Cherubim et al. (2026)] (hereafter C26) is interpreted as a wind with $\text{H}:\text{He} \lesssim 10^{-3}$, that is, a gas in which hydrogen is a trace species. Every EXHALE configuration validated to date is hydrogen-dominated, and in all of them Lyman- α is the dominant line coolant. Removing hydrogen therefore removes the coolant the code has always relied on, and the question of what replaces it must be answered before a helium-rich wind solution can be trusted. This is the energy-closure item of Phase C in `lhs1140b_lower_atmosphere_plan_new.md`.

2. Line channels available to helium

2.1 What the code implements

The collisional-excitation cooling terms are assembled in `util_ion_eq.f90` and their coefficients defined in `Cool_coeff.f90`. Reading the exponential threshold of each coefficient identifies the transition it represents:

Coefficient	Threshold	Transition	Routine
$7.5 \times 10^{-19} (1 + \sqrt{T/10^5})^{-1} e^{-118348/T}$	10.20 eV	H I Ly α 1216 Å	<code>coex_rate_HI</code>
$1.1 \times 10^{-19} T^{0.082} e^{-2.3 \times 10^5/T}$	19.82 eV	He I $1^1S \rightarrow 2^3S$	<code>coex_rate_HeI</code>
$5.54 \times 10^{-17} T^{-0.397} (1 + \sqrt{T/10^5})^{-1} e^{-473638/T}$	40.81 eV	He II Ly α 304 Å	<code>coex_rate_HeII</code>
$1.16 \times 10^{-20} \sqrt{T} e^{-13179/T}$	1.14 eV	He I $2^3S \rightarrow 2^3P$, 10830 Å	<code>coex_rate_HeI23S_10830</code>

Table 1: Collisional-excitation cooling coefficients (cgs, to be multiplied by n_e and the population of the lower level). The threshold energies are the exponential arguments converted with k_B ; each matches a known level to three digits, which is how the transitions above were identified rather than assumed.

Two further terms act on the metastable population when the triplet is tracked: the collisional conversions $2^3S \rightarrow 2^1S$ (0.80 eV) and $2^3S \rightarrow 2^1P$ (1.40 eV), through `coex_HeI_23S_21S` and `coex_HeI_23S_21P`. The ground-to-triplet excitation q_{13} is deliberately excluded there because the 19.82 eV term in the table already includes it; the code comment records this to prevent counting it twice.

The 10830 Å channel and the two conversions scale with the explicitly computed $n(2^3S)$. That population is supplied by recombination and is small, so these channels track a minority level and cannot act as a bulk thermostat, however low their thresholds are.

2.2 What the single exponential actually represents

The 584 Å resonance line ($1^1S \rightarrow 2^1P$, 21.22 eV) is the optically allowed counterpart of Lyman- α and has no term of its own in `Cool_coeff.f90`. Reading the thresholds alone, one would conclude that the code includes only the 19.82 eV metastable channel and leaves out the resonance line. Checking that conclusion against CHIANTI shows it is wrong, in both directions, and the correction matters enough to state explicitly.

Summing every transition out of the He I ground level in CHIANTI v11 (`cooling_data/fit_helium_cooling.py`) gives the following. First, the 584 Å line is *not* the dominant channel at wind temperatures: relative to the 2^3S channel it contributes 1% at 5000 K, 6% at 10^4 K and 52% at 5×10^4 K, because its 1.4 eV higher threshold costs more than its larger collision strength returns until the gas is very hot. Second, and more usefully, the existing coefficient is not a fit to the 2^3S line at all: it reproduces the *total* ground-state sum to within 5–19% over 5×10^3 – 5×10^4 K, while departing from the 2^3S channel alone by up to a factor 3.4. Its 19.82 eV exponent gives it the temperature dependence of the lowest channel, and its prefactor is large enough to match the sum of all of them.

The resonance line is therefore already included in the total this formula was fitted to, so the practical question is how accurate that formula is, not whether the line is present. Section 5. refits both helium coefficients to CHIANTI and reports what that is worth; the larger error turns out to be in He II, not He I. Those fits are recorded there, not installed: `Cool_coeff.f90` still evaluates the two expressions of Table 1.

3. How weak the helium channels are

Evaluating the coefficients of Table 1 directly:

T [K]	H I Ly α	He I 19.8 eV	He II 304 Å	He I/H I
5000	3.22×10^{-29}	2.33×10^{-39}	1.12×10^{-59}	7.2×10^{-11}
8000	2.20×10^{-25}	7.51×10^{-32}	2.36×10^{-44}	3.4×10^{-7}
10000	4.13×10^{-24}	2.40×10^{-29}	2.93×10^{-39}	5.8×10^{-6}
15000	2.02×10^{-22}	5.30×10^{-26}	1.70×10^{-32}	2.6×10^{-4}
20000	1.40×10^{-21}	2.51×10^{-24}	3.90×10^{-29}	1.8×10^{-3}
30000	9.38×10^{-21}	1.20×10^{-22}	8.31×10^{-26}	1.3×10^{-2}
50000	4.12×10^{-20}	2.69×10^{-21}	3.40×10^{-23}	6.5×10^{-2}

Table 2: Collisional-excitation cooling coefficients in cgs, evaluated from the implemented expressions. The last column is the ratio of the He I channel to Lyman- α at equal lower-level density.

The comparison is per atom of the relevant species, so in a wind where helium outnumbered hydrogen by 10^3 the ratio in the last column is raised by that factor; even then the helium channel only reaches parity with what a trace

hydrogen population would supply near 10^4 K, and falls behind at lower temperature. The He II line is irrelevant below $\sim 3 \times 10^4$ K: at 40.81 eV it is essentially extinct in a photoionized wind at a few thousand kelvin.

4. What closes the energy budget instead

With lines suppressed, the terms that survive are the continuum ones already in `util_ion_eq.f90` — recombination cooling of H II/He II/He III, collisional-ionization cooling, and free-free emission — together with the PdV work of the expanding wind, which is not a radiative term at all but is what actually carries the energy away in a fast outflow. The first converged case of the Phase C sweep already shows the transition. At He/H = 1 (the tutorial planet with only the helium ratio changed), the channel-resolved output gives heating led by He I (96% of the total at the base) and cooling led by recombination (75% at the base, 91% at the top); Lyman- α leads only in the intermediate layer at $1.5 R_p$ (54%, $T = 1.6 \times 10^4$ K), where hydrogen is still half of the nuclei. The remaining cases of the sweep (He/H = 10, 100, 1000) test whether that last Lyman- α peak disappears and the budget closes on the continuum channels alone, which is the acceptance criterion for this phase.

5. CHIANTI-based replacement coefficients

Both helium coefficients were refitted to CHIANTI v11 with the same method already used for the metal coolants (`cooling_data/chianti_cooling.py` reads the raw `.elvlc/.wgfa/.scups` files and descales the effective collision strengths following Burgess & Tully 1992; no ChiantiPy). The fitted quantity is the sum over all transitions out of the ground level, which is what the existing coefficients represent, so the rule of Section 2.1 is unchanged: the ground-to-triplet channel stays inside this term, and is still left out of the triplet system so that it is not counted twice.

The functional form is the one `fit_cooling_formulas.py` uses, exact for a single line and a good effective form for a sum whose shape is set by its lowest threshold,

$$\Lambda(T) = \frac{C}{\sqrt{T}} \left[a + b \ln \left(1 + \frac{T}{T_0} \right) \right] e^{-T_{\text{ex}}/T}, \quad C = \frac{8.629 \times 10^{-6} \Delta E}{g_l}, \quad T_{\text{ex}} = \frac{\Delta E}{k_B}, \quad (1)$$

with ΔE left free so the fit recovers the effective threshold of the sum rather than assuming the lowest level. It recovers 19.88 eV for He I and 40.76 eV for He II, both within 0.3% of the lowest level of each ion, which is the expected behavior for a threshold-dominated sum.

Ion	g_l	C	a	b	T_0 [K]	T_{ex} [K]
He I	1	2.749039×10^{-16}	0.0613017	0.4191808	9.821387×10^4	230747.5
He II	2	2.817309×10^{-16}	0.4371421	0.6051062	1.905585×10^5	472956.0

Table 3: Fitted coefficients, cgs. Fit ranges 3×10^3 – 10^5 K (He I) and 10^4 – 2×10^5 K (He II).

Table 4 gives the accuracy against the CHIANTI sum, together with the error of the coefficients these would replace. The He I coefficient is therefore adequate at the tens-of-percent level and biased low; the He II coefficient is low by a factor of two and degrades with temperature, so it is the one worth replacing on physical grounds. Neither error is

	this fit	current Cool_coeff.f90
He I, 5×10^3 – 5×10^4 K	0.4%	–19.4% (ratio 0.81–0.95)
He II, 2×10^4 – 10^5 K	0.2%	–53.6% (ratio 0.46–0.73)

Table 4: Maximum deviation from the CHIANTI ground-state sum over the range where each coolant can matter.

important in a hydrogen-rich run, where Lyman- α exceeds both by orders of magnitude, which is presumably why they survived; both act directly on the thermal structure of a helium-dominated wind. Fitting scripts, coefficient file (`helium_cooling_fits.txt`) and comparison figure (`helium_cooling_fits.pdf`) are in `cooling_data/`. The coefficients are recorded here but not yet installed in the Fortran; doing so changes converged solutions and therefore needs its own golden refresh.

A caveat that applies to both, and to the coefficients they replace: this is the optically thin, low-density (coronal) limit, in which every collisional excitation is followed by a radiative decay that escapes. The 584 Å line is a resonance line of the dominant species in a helium-rich wind, so it can be optically thick, and the 2^3S channel feeds a metastable level that EXHALE tracks explicitly when the triplet is on. Both effects reduce the true cooling below the coronal value; the fits above bound it from that side rather than resolving it.

6. Consequence: a wind without a thermostat

Line cooling is what keeps a hydrogen-rich thermosphere near 10^4 K: Lyman- α switches on sharply once the gas reaches the excitation threshold and absorbs further heating at nearly constant temperature. A helium-dominated gas has no such switch below several times 10^4 K. It should therefore run hotter for the same heating rate, and its temperature should be set by the balance between photoionization heating, recombination, and expansion rather than pinned by a line.

This is consistent with what the C26 retrieval requires ($T_{\text{wind}} \simeq 5160$ K, with their grid explored to 6400 K) and with our own p-winds oracle, in which temperature is by far the strongest lever on the line depth while the mass-loss rate is weak and saturated (`LHS1140b/pwinds_oracle/README.md`). It is a consistency argument, not yet a demonstration: only a solution with a computed energy equation can show that such a temperature is actually maintained.

7. What C26 says about cooling

Nothing. The paper contains no radiative-cooling discussion, no energy equation, and no heating efficiency. This is structural rather than an oversight: p-winds models the outflow as an *isothermal* Parker wind, so the outflow temperature is an input. The Supplement states it plainly: the code forward models the spectrum “for given values of the atmospheric escape rate, outflow temperature, H:He atomic number ratio, line-of-sight wind speed, top-of-atmosphere XUV spectrum, planetary mass and radius”, and T_{wind} is then estimated by MCMC alongside the other three parameters. The stellar spectrum enters only through the ionization rates and the resulting number densities, never as a heating rate.

A full-text search of the article and its Supplement finds the word “cool” in three places, none of them about cooling physics: “cooler, rocky exoplanets” in the introduction, “the cool equilibrium temperature” in the cold-trap discussion, and a reference title. “Isothermal” appears twice: once for the wind and once for the APEC model of the stellar corona, which concerns the X-ray fit and not the planet. No heating efficiency η is quoted anywhere.

The two places where energetics do appear are global budget arguments, not statements about the interior of the wind:

1. an energy-limited estimate of $(6.2\text{--}29) \times 10^7 \text{ g s}^{-1}$, obtained by integrating the scaled SEDs over 10–1300 Å, which brackets their retrieved $\dot{M} = 2.03^{+0.58}_{-0.67} \times 10^8 \text{ g s}^{-1}$;
2. the exclusion of core-powered mass loss on the grounds that it predicts a much smaller rate.

A related check they do report is a composition test rather than an energy one: fixing the mean molecular weight at 4 amu, the theoretical maximum for a pure neutral helium wind, leaves the transmission spectra unchanged.

8. Implication for the extension

The retrieved T_{wind} is an observational constraint on the line profile and the metastable population, not a temperature shown to be thermodynamically sustainable. Supplying that missing half is precisely the value of running EXHALE on this target: EXHALE computes the temperature from a heating–cooling balance, so the question “does a helium-dominated wind settle near 5000–6400 K under this irradiation?” becomes answerable rather than assumed. The weakness of helium line cooling documented above makes a hot, poorly regulated solution the physically expected outcome, and makes the accuracy of the few surviving channels — the He II coefficient above all, low by a factor of two — more consequential than it would be in any hydrogen-rich run.

References

[Cherubim et al. (2026)] C. Cherubim et al., *Helium escaping from the atmosphere of a nearby rocky exoplanet orbiting in a habitable zone*, Science, 10.1126/science.aea9708 (2026); first release 2026 July 16, with Supplementary Materials.